How to work a question

Every answer in this key is laid out the same way, so you see the path, not just the answer.

1. **Find the key words.** Before you think about the answer, circle the words that tell you *which decision* you're making: rack type, flue, commodity class, density, hose stream. Those words pick the chapter. The numbers in the question (heights, depths, aisle widths) are what you test against the rule.
2. **Go where it lives.** Chapter, then section, then table, with the page number. In the key this is **WHERE I WENT**. On "Say it" questions it tells you where the rule is, so you can look it up afterward.
3. **Test your numbers against the rule.** Check every condition, one at a time. Don't stop at the first one that fits. In the key this is **HOW I GOT THE ANSWER**.
4. **Check the trap.** Ask what the common wrong answer is and why someone lands on it. In the key this is **THE TRAP**. Most misses are a skipped condition, not a wrong number.

WORKED EXAMPLE: EPISODE 3, QUESTION 4

> *Two double-row racks, each eight feet deep, with a three-foot aisle between them. What rack type?*
>
> **1 · Key words:** "what rack type" is the decision. The numbers to test: **8 ft** deep, **3 ft** aisle, **8 ft** deep.
>
> **2 · Where it lives:** rack types are defined in Chapter 3: single-row, double-row, multiple-row.
>
> **3 · Test the numbers:** each rack is 8 ft, under 12 ft, so it *looks* double-row. But there's a second condition: the aisle. 3 ft is under 3½ ft, so the aisle doesn't separate them. Add it all up: 8 + 3 + 8 = 19 ft, over 12 ft. **Multiple-row.**
>
> **4 · The trap:** stopping after the depth check and answering double-row. The aisle is the condition people skip.

How to study with it: cover the key under a question. Answer out loud *and say your path* ("rack type, Chapter 3, aisle under 3½, so…"). Then uncover one line at a time. Saying the path is what builds the habit.

Citations are written as numbers here (§20.5.3.4, p. 13-184). Page numbers are from the printed 2022 edition. The facts are the same ones the audio uses; confirm them in your book.

FSM STORAGE PROCEDURE · NFPA 13 (2022) · ANSWER KEY

Episode 1: The Route Map

6 answers. Cover everything under a question, say your answer and your path out loud, then uncover one line at a time.

STEP ZERO. THE HIGH-PILED GATE.

1. SAY IT · NO BOOK **A Class Three commodity, stored ten feet high. Under the 2022 edition, is it high-piled or low-piled, and where do you go next?**

THE ANSWER

Low-piled. You go to Chapter 4, the low-piled section.

WHERE I WENT

The high-piled definition sets the line, and the low-piled section in Chapter 4 protects it.

HOW I GOT THE ANSWER

Class One through Class Four is high-piled only at more than twelve feet. Ten is not more than twelve.

THE TRAP

If you said high-piled: you used the five-foot line. That line is only for Group A plastics. A Class One through Four commodity still uses twelve.

STEP TWO. CHAPTER TWENTY, GENERAL REQUIREMENTS FOR STORAGE.

2. FIND IT · BOOK OPEN **A customer adds a building of baled cotton. Which section in Chapter 20 covers baled cotton, and where does it send you for protection?**

THE ANSWER

§20.4.11.1, the baled cotton section. It sends you to Chapter 26, §26.5.

WHERE I WENT

Chapter 20, Commodity Classes, p. 13-183. Look in the right column.

HOW I GOT THE ANSWER

Baled cotton is a special commodity, and special commodities are classified in 20.4, Commodity Classes. The section number tells you what it is. The section it points to tells you where it's protected.

THE TRAP

If you said 20.4.11: you stopped one level too high. 20.4.11 is Display and Storage of Up to Group A Plastics, and it sends you to 26.3. Baled cotton is its point one.

---

3. SAY IT · NO BOOK **Name the three things you write down before you leave Chapter 20.**

THE ANSWER

The commodity class, the storage arrangement, and the storage height.

WHERE I WENT

Chapter 20. Commodity classes at 20.4, storage arrangement at 20.5, heights at 20.9.

HOW I GOT THE ANSWER
Those three are your keys for every table that comes after.

THE TRAP
If you said the sprinkler density: Chapter 20 doesn't give you the sprinkler density yet. That comes from the protection chapter you pick in step four.

STEP THREE. SCREEN THE BUILDING.

4. SAY IT · NO BOOK **The roof slopes three inches in twelve. Which sprinkler technology does that take off the table, and which rule says so?**

THE ANSWER
ESFR.

WHERE I WENT
§14.2.3, the ESFR slope limit. Two in twelve, maximum. The storage criteria in 20.9.1 are written for that same two in twelve.

HOW I GOT THE ANSWER
Three in twelve is steeper than two in twelve, and for ESFR that's a hard limit. And the other storage criteria are written for two in twelve too, so a steeper roof needs a section that specifically allows it.

THE TRAP
If you said you'd find out in Chapter 23: the slope limit is in Chapter 14. That's why you screen the building before you open the protection chapter.

STEP FOUR. PICK THE PROTECTION CHAPTER.

5. SAY IT · NO BOOK **You've picked your protection chapter, and its table doesn't cover your commodity, arrangement, and height. What are your two moves?**

THE ANSWER
Move to another chapter, or, if it's rack storage, add in-rack sprinklers under Chapter 25.

WHERE I WENT
Chapter 25 is In-Rack Sprinklers, for when the ceiling alone can't do the job.

HOW I GOT THE ANSWER
The table only covers what it lists. You design from inside a table, never past it.

THE TRAP
If you said use the closest row that doesn't quite cover you: that's stretching the table, and you don't stretch it.

STEP FIVE. HOSE ALLOWANCE AND DURATION.

6. FIND IT · BOOK OPEN **Your warehouse is ESFR. Where do you find the hose stream**

allowance and the water supply duration?

> **THE ANSWER**
>
> **Table 20.15.2.6, the hose table.**

> **WHERE I WENT**
>
> Chapter 20, §20.15, Hose Stream Allowance and Water Supply Duration. p. 13-190. Find your commodity, then your sprinkler type, then the row for the number of sprinklers in your design. The table has rows for CMDA, CMSA, and ESFR.

> **HOW I GOT THE ANSWER**
>
> The words hose and duration send you to Chapter 20, §20.15. The table has an ESFR row, so you read that one. If you went looking in Chapter 23: that's where the ESFR sprinkler design is, but hose and duration come back to this one table in Chapter 20.

Episode 2: Classify the Commodity

9 answers. Cover everything under a question, say your answer and your path out loud, then uncover one line at a time.

WARM-UP

1. SAY IT · NO BOOK **Group A plastic, stored eight feet high. Under the 2022 edition, is that high-piled storage?**

THE ANSWER

Yes. Eight is more than five.

WHERE I WENT

Chapter 3, §3.3.100, High-Piled Storage.

HOW I GOT THE ANSWER

2022 has two lines. More than twelve feet for Class One through Four. More than five feet for Group A plastics.

THE TRAP

If you said no: you used the twelve-foot line, the 2019 rule for everything.

---

2. SAY IT · NO BOOK **You need the hose stream allowance and the water supply duration for a storage design. Which table?**

THE ANSWER

Table 20.15.2.6.

WHERE I WENT

Chapter 20, p. 13-190. The last stop on the map.

---

3. SAY IT · NO BOOK **Rubber tires in storage. Do their design criteria come from Chapter 26?**

THE ANSWER

No.

WHERE I WENT

Tires are classified in Chapter 20, §20.4.9. Their design criteria sit inside Chapters Twenty-One, Twenty-Two, and Twenty-Three, one section each.

THE TRAP

If you said yes: you lumped tires in with baled cotton. Of tires, roll paper, and baled cotton, only baled cotton goes to Twenty-Six.

STEP A. FIND THE PLASTIC.

4. FIND IT · BOOK OPEN **A pallet load of products made of silicone rubber. What is it**

protected as?

THE ANSWER

Class Four.

WHERE I WENT

Chapter 20, §20.4.6, the Group B list, p. 13-183. Silicone rubber is on it. Then §20.4.8.1, on the same page: Group B is protected the same as Class Four.

HOW I GOT THE ANSWER

The question named a material. A material name sends you to the plastics lists, not the class ladder. Find the group, then read the rule for that group.

THE TRAP

If you said Class Three: you read it as Group C. Silicone rubber is on the Group B list, and Group B rides as Class Four.

STEP B. THE CLASS ONE THROUGH FOUR LADDER.

5. SAY IT · NO BOOK **A cartoned product that's ten percent nonexpanded Group A plastic by weight. What class is it?**

THE ANSWER

Class Four.

WHERE I WENT

§20.4.4, Class Four, p. 13-181.

HOW I GOT THE ANSWER

Cartoned nonexpanded, more than five and up to fifteen percent by weight. The middle band is Class Four.

THE TRAP

If you said Class Three: Class Three allows only five percent or less. If you said Group A nonexpanded: cartoned needs more than fifteen.

STEP C. GROUP A: CARTONED OR EXPOSED, EXPANDED OR NONEXPANDED.

6. FIND IT · BOOK OPEN **An exposed product, no carton. Its only plastic is expanded Group A, at twenty percent by volume. Is it Class Four, or something else?**

THE ANSWER

Group A nonexpanded.

WHERE I WENT

Chapter 20, §20.4.5.4, the nonexpanded definition, p. 13-183. Read the exposed line written by volume: more than five, up to twenty-five percent expanded.

HOW I GOT THE ANSWER

Two words picked it, exposed and expanded. The Class Four list has no exposed expanded line, so the answer has to be in the Group A definitions.

THE TRAP

If you said Class Four: you used the cartoned line, more than five up to twenty-five by volume. That line needs a carton.

STEP D. CHECK THE PALLET.

7. FIND IT · BOOK OPEN **A cartoned Class Three product on unreinforced H-D-P-E pallets, permanently marked. The building has ceiling-only spray sprinklers, K-16.8. Does the commodity class go up?**

THE ANSWER

No. It stays Class Three.

WHERE I WENT

Chapter 20, the K-16.8 exception, p. 13-181. The increase rule is on p. 13-180. Turn the page for the exception.

HOW I GOT THE ANSWER

H-D-P-E pallet sent you to Pallet Types. Ceiling-only spray, K-16.8, sent you to the exception. It's keyed on the pallet material and the sprinkler.

THE TRAP

If you said Class Four: you applied the unreinforced-pallet rule and stopped on p. 13-180.

STEP E. CHECK FOR ENCAPSULATION.

8. SAY IT · NO BOOK **Palletized storage, encapsulated, Class Three, eighteen feet high. What's the minimum K-factor, and what's the density?**

THE ANSWER

K-11.2 or larger, and point six over two thousand square feet.

WHERE I WENT

Chapter 21, §21.2.3, starting on p. 13-195.

HOW I GOT THE ANSWER

Encapsulated, over fifteen feet up to twenty, palletized. Eighteen is inside that range.

THE TRAP

If you said point five three: that's Class Two. Class Three and Four share point six.

STEP F. CHECK FOR MIXED COMMODITIES.

9. FIND IT · BOOK OPEN **A building whose ceiling protection is designed for Class Two. The owner wants to keep some Group A pallet loads scattered through a thirty-thousand-square-foot area. How many loads are allowed?**

THE ANSWER

Five pallet loads, scattered, with no two adjacent in any direction, diagonals included.

WHERE I WENT

Chapter 20, the ten-load allowance, p. 13-184. Read the conditions under it, down to the ceiling design.

HOW I GOT THE ANSWER

A few loads of a worse commodity is a mixed-commodity question. The allowance is keyed on the area and on what the ceiling is designed for.

THE TRAP

If you said ten: you missed the condition. A Class One or Two ceiling drops it to five.

Episode 3: Arrangement and Rack Type

9 answers. Cover everything under a question, say your answer and your path out loud, then uncover one line at a time.

WARM-UP

1. SAY IT · NO BOOK **Group B plastic, and free-flowing Group A plastic. What are they protected the same as?**

THE ANSWER

Class Four. Both of them.

WHERE I WENT

§20.4.8.1, p. 13-183.

HOW I GOT THE ANSWER

Free-flowing plastics fall out of their containers in a fire, fill the flue spaces, and smother the fire. That's why they ride down to Class Four. Group B is simply listed in Class Four's own list, item one.

THE TRAP

If you said Class Three: you swapped the groups. Group C is the one protected the same as Class Three, 20.4.8.2.

---

2. SAY IT · NO BOOK **An exposed product, Group A plastic, with more than five, up to twenty-five percent by volume of expanded plastic. What is it?**

THE ANSWER

Group A nonexpanded.

WHERE I WENT

§20.4.5.4, the nonexpanded definition, p. 13-183.

HOW I GOT THE ANSWER

Class Four has no line for exposed expanded plastic. So the exposed product skips Class Four and lands in the plastics groups.

THE TRAP

If you said Class Four: you used the cartoned line. Cartoned, more than five up to twenty-five percent by volume expanded is Class Four. Exposed is not. The carton is worth a whole category.

---

3. SAY IT · NO BOOK **A Class Two product on polypropylene pallets with no marking, under K-11.2 ceiling sprinklers. What class do you protect it as?**

THE ANSWER

Class Four.

WHERE I WENT

§20.3.2, Pallet Types, p. 13-180.

HOW I GOT THE ANSWER

No marking and no manufacturer certification means you assume reinforced. Reinforced raises it two classes. Class Two becomes Class Four.

THE TRAP

If you said Class Three: you used the unreinforced rule, up one class. That only applies when the pallet is permanently marked as unreinforced.

STEP B. NAME THE RACK TYPE.

4. SAY IT · NO BOOK **Two double-row racks, each eight feet deep, with a three-foot aisle between them. What rack type?**

THE ANSWER

Multiple-row.

WHERE I WENT

Chapter 3, the multiple-row rack definition, §3.3.134.

HOW I GOT THE ANSWER

Three feet is under three and a half, so the aisle doesn't count. Eight, plus three, plus eight is nineteen. Over twelve, with an aisle under three and a half. Under three and a half feet of aisle, two racks become one.

THE TRAP

If you said double-row: you checked each rack's depth, saw eight is under twelve, and stopped. You never checked the aisle. Check the aisle every time.

---

5. SAY IT · NO BOOK **A single-row rack is set eighteen inches from a wall. What rack type is it treated as, and where is its longitudinal flue?**

THE ANSWER

Double-row. The longitudinal flue is the gap between the rack and the wall.

WHERE I WENT

The wall rule, in the annex, A.3.3.200.

HOW I GOT THE ANSWER

Eighteen inches is within twenty-four inches of the wall. The wall plays the missing back row.

THE TRAP

If you said single-row: you read only the Chapter 3 definition and missed the annex. If you said the wall is the flue: the wall is not the flue. The gap is.

STEP C. CHECK THE FLUES.

6. FIND IT · BOOK OPEN **A double-row open rack holds a Class Three commodity, stored to twenty-two feet. Is a longitudinal flue required?**

THE ANSWER

No. Not required.

WHERE I WENT

Chapter 20, §20.5.3.4, p. 13-184. Find the longitudinal flue paragraphs, point four point one, and read the first one, point one point one, the twenty-five-and-under flue rule.

HOW I GOT THE ANSWER

The words longitudinal flue and double-row open rack picked the flue section. The rule is keyed on storage height, with twenty-five feet as the line, and on the commodity, Class One through Four or Group A. Class Three at twenty-two feet is inside it.

THE TRAP

If you said yes, six inches: you read the over-twenty-five flue rule. That one starts over twenty-five feet. Twenty-two is under. And remember, not required doesn't mean the rack stays open on its own. That's Step D.

STEP D. OPEN RACK OR SOLID SHELF.

7. FIND IT · BOOK OPEN **A double-row rack has no shelves. Storage is twenty-four feet. The operator pushed the loads back to back, so there's no longitudinal flue. Transverse flues are four feet apart. Open rack or solid-shelf rack?**

THE ANSWER

Open rack.

WHERE I WENT

Chapter 20, §20.5.3.1, p. 13-184. Find the double-row paragraphs, point one point two. Point two point one, the no-flue rule, makes it solid. Keep reading to point two point two, the way-out rule.

HOW I GOT THE ANSWER

The question is open or solid, so you're in the shelving section, not the flue section. Double-row picks point one point two. The way-out rule is keyed on two things: storage twenty-five feet or less, and transverse flues no more than five feet apart. Twenty-four feet and four feet pass both.

THE TRAP

If you said solid: you stopped at the no-flue rule and never read the way out. Always read the next paragraph. Now push the same rack to thirty feet, and the way out is gone. That's our forklift drill.

---

8. FIND IT · BOOK OPEN **A double-row rack has one solid steel shelf, fifteen square feet in area. The rest of the rack is open, with its six-inch longitudinal flue and transverse flues in place. Does that shelf make it a solid-shelf rack?**

THE ANSWER

No.

WHERE I WENT

Chapter 20, §20.5.3.1.1, the solid shelf test, p. 13-184, left column.

HOW I GOT THE ANSWER

The words solid shelf picked the solid shelf test. It's keyed on two things: how open the shelving is, and the area. The area has to be greater than twenty square feet. Fifteen isn't.

THE TRAP

If you said yes: you hung the twenty square feet on the blocked-flue half only. Read the sentence again. The twenty applies to both halves.

STEP E. THE HEIGHTS.

9. SAY IT · NO BOOK **Most of the year a building stores to twenty-two feet. Every peak season, the same racks are stacked to twenty-eight feet. What storage height do you design for?**

THE ANSWER

Twenty-eight feet.

WHERE I WENT

The design height rule, §20.9.3.1.

HOW I GOT THE ANSWER

Twenty-eight feet exists periodically, every peak season, and the higher stack creates the greater water demand. Both halves of the rule point at twenty-eight.

THE TRAP

If you said twenty-two feet: you designed to today's walkthrough. Routinely or periodically. The peak season counts.

Episode 4: CMDA

9 answers. Cover everything under a question, say your answer and your path out loud, then uncover one line at a time.

WARM-UP

1. SAY IT · NO BOOK **Two ten-foot double-row racks stand with a three-foot aisle between them. What rack type does the code see?**

THE ANSWER

Multiple-row.

WHERE I WENT

The multiple-row rack definition in Chapter 3, §3.3.134.

HOW I GOT THE ANSWER

Three feet is under three and a half, so the aisle doesn't count as an aisle anymore. The code sees one rack twenty-three feet wide. Over twelve is multiple-row.

THE TRAP

If you said double-row: you checked each rack's depth and skipped the aisle. Under three and a half feet of aisle, two racks become one.

---

2. SAY IT · NO BOOK **Double-row open racks. Above what storage height is a six-inch longitudinal flue required?**

THE ANSWER

Over twenty-five feet.

WHERE I WENT

§20.5.3.4.1.2, p. 13-184.

HOW I GOT THE ANSWER

At twenty-five feet or less, the section just before it, 20.5.3.4.1.1, says a longitudinal flue is not required. Over twenty-five, a nominal six-inch one is.

THE TRAP

If you said twenty-five feet: at exactly twenty-five it isn't required. The line is over twenty-five.

---

3. SAY IT · NO BOOK **A thirty-foot double-row rack with no shelves in it at all. The owner pushes the loads back to back and closes the longitudinal flue. Open rack, or solid shelf?**

THE ANSWER

It's treated as a solid-shelf rack.

WHERE I WENT

§20.5.3.1.2.1, p. 13-184.

HOW I GOT THE ANSWER

A double-row rack with no longitudinal flue is treated as solid-shelf. The one way out needs storage of twenty-five feet or less, with transverse flues no more than five feet apart. Thirty feet doesn't qualify.

THE TRAP

If you said open, because there are no shelves: missing flues turn an open rack into a solid-shelf rack, even with no shelf in it.

STEP A. FIND YOUR SECTION.

4. SAY IT · NO BOOK **Class Three commodity in double-row racks, thirty feet of storage, CMDA. Where does Chapter 21 send you, and what comes with it?**

THE ANSWER

Chapter 25. Ceiling sprinklers plus in-rack sprinklers.

WHERE I WENT

§21.4.2.1, p. 13-207.

HOW I GOT THE ANSWER

Class One through Four, over twenty-five feet. That's §21.4.2, and its first rule hands you to Chapter 25, the in-rack chapter.

THE TRAP

If you said a density table in §21.4.1: those tables stop at twenty-five feet. Thirty is over.

STEP B. READ THE TABLE.

5. FIND IT · BOOK OPEN **Double-row racks, Class Two, not encapsulated, twenty-four feet of storage, eight-foot aisles, ordinary-temperature sprinklers. What density does the table give?**

THE ANSWER

Point six five gallons per minute per square foot.

WHERE I WENT

Chapter 21, Table E. The page number isn't on our photo, so look for it ahead of p. 13-207. Go down to Class Two, not encapsulated, take the eight-foot aisle line, and read the last column, ordinary-temperature.

HOW I GOT THE ANSWER

Double-row picks the single- and double-row tables, not the multiple-row ones. Twenty-four feet picks part E, over twenty-two up to and including twenty-five. Then the table is keyed on commodity class, encapsulation, aisle width, and temperature rating, in that order across the row.

THE TRAP

If you said point five eight: you read the high-temperature column. That's the lower number. If you said point seven seven: you read the four-foot aisle line. And if you said "See Chapter 25": you were in the multiple-row table. Remember, that's a density only. Table E doesn't give a design area.

---

6. SAY IT · NO BOOK **Twenty-four feet of storage, not encapsulated. Class Four in double-row racks, and Class Two in multiple-row racks sixteen feet deep with eight-foot aisles. Which one can CMDA protect with ceiling sprinklers only?**

THE ANSWER

Neither one.

WHERE I WENT

Table E for the double-row rack, and multiple-row Table E on p. 13-207.

HOW I GOT THE ANSWER

In Table E, Class Four not encapsulated reads in-racks mandatory, yes. In multiple-row Table E, only Class One not encapsulated gets a ceiling-only density. Class Two says "See Chapter 25."

THE TRAP

If you said the Class Two rack: you carried the double-row pattern into the multiple-row table. Only Class One gets through there.

STEP C. THE ADJUSTMENTS.

7. SAY IT · NO BOOK **The encapsulation adder. What rack type and height band does it cover, and what does it add for Class One through Three, and for Class Four?**

THE ANSWER

Multiple-row racks, over twelve feet up to and including twenty-five. Class One through Three, twenty-five percent. Class Four, fifty percent.

WHERE I WENT

§21.4.1.6, p. 13-207.

HOW I GOT THE ANSWER

Encapsulation is the trigger and multiple-row is the rack type. Over twenty-five feet a different rule takes over, §21.4.2.2, and it adds twenty-five percent for encapsulated Class One through Four.

THE TRAP

If you said twenty percent: that's the slave pallet rule, and it drops off when in-racks are in the design.

STEP D. RUN OUR WAREHOUSE.

8. FIND IT · BOOK OPEN **Cartoned Group A plastic, thirty feet of storage, and the designer**

puts the top in-rack level at twenty-two feet. What ceiling design do you use?

THE ANSWER

Point four five gallons per minute per square foot, over two thousand square feet.

WHERE I WENT

Chapter 25, the cartoned plastics section, then the option figures. Start with part A, option one. The page number isn't on our photo. Read the ceiling design printed across the top of the figure.

HOW I GOT THE ANSWER

Group A plastic, over twenty-five feet. §21.6 sends you to Chapter 25, and cartoned picks the cartoned plastics section. The ceiling line is keyed on the storage above the top in-rack level, not on the total storage height. Thirty minus twenty-two is eight feet. Over five, up to ten.

THE TRAP

If you said point three zero: you took the five-feet-or-less line. That needs the top level at twenty-five feet or higher. And if the designer had put the top level at eighteen feet, that's twelve feet above it. The ten-foot rule says no.

STEP E. HOSE AND DURATION.

9. FIND IT · BOOK OPEN **A different job, CMDA, with a ceiling design area of fourteen hundred square feet. What hose stream and duration?**

THE ANSWER

Five hundred gallons per minute, ninety minutes.

WHERE I WENT

Chapter 20, the hose table, p. 13-190. Find the CMDA line, then read down the column for size of design area at ceiling.

HOW I GOT THE ANSWER

Hose and duration for this design come from this table. For CMDA, the rows are keyed to the size of the design area. The CMSA and ESFR rows in the same table are keyed to the number of sprinklers instead.

THE TRAP

If you said a hundred twenty minutes: that's the over-fifteen-hundred row, the one our two-thousand square foot warehouse uses. If you said two hundred fifty gallons per minute and sixty minutes: that row stops at twelve hundred.

Episode 5: CMSA and ESFR

9 answers. Cover everything under a question, say your answer and your path out loud, then uncover one line at a time.

WARM-UP

1. SAY IT · NO BOOK **Our warehouse under CMDA. Cartoned Group A plastic, thirty feet of storage. Can the ceiling sprinklers do it alone?**

THE ANSWER

No. Over twenty-five feet, CMDA never stands alone. In-racks, every time.

WHERE I WENT

§21.6, p. 13-207. It sends you to Chapter 25, the in-rack chapter.

HOW I GOT THE ANSWER

Over twenty-five feet, both CMDA paths leave Chapter 21. Class One through Four goes by §21.4.2.1, Group A plastic goes by §21.6, and both land in Chapter 25.

THE TRAP

If you said yes, with a higher density: you stayed in Chapter 21. Over twenty-five feet, §21.6 sends you out to Chapter 25.

---

2. SAY IT · NO BOOK **Same warehouse, CMDA with in-racks. The top in-rack level sits at twenty-five feet, so there are five feet of storage above it. What's the ceiling design?**

THE ANSWER

Point three zero gallons per minute per square foot, over two thousand square feet.

WHERE I WENT

The ceiling design across the top of Figures 25.3.3.3.1.1, parts A through G. Call them the seven option figures.

HOW I GOT THE ANSWER

Five feet or less above the top in-rack level gets point three zero. Over five feet, up to ten feet, gets point four five.

THE TRAP

If you said point four five: you put five feet in the upper band. Point four five starts over five feet.

---

3. SAY IT · NO BOOK **That CMDA ceiling design covers two thousand square feet. What's the hose allowance and duration?**

THE ANSWER

Five hundred gallons per minute, a hundred twenty minutes.

WHERE I WENT

Table 20.15.2.6, p. 13-190. Call it the hose table.

HOW I GOT THE ANSWER

For CMDA, the hose table is keyed on the size of the design area. Two thousand falls in over fifteen hundred, up to twenty-six hundred.

THE TRAP

If you said ninety minutes: you read the over twelve hundred, up to fifteen hundred row. Two thousand is past that.

STEP A. SCREEN THE BUILDING.

4. SAY IT · NO BOOK **An ESFR area sits right next to an area of standard-response sprinklers. What has to go between them, and how big?**

THE ANSWER

A noncombustible draft curtain at least two feet deep, with a clear aisle at least four feet wide centered under it.

WHERE I WENT

§23.1.2, p. 13-225.

HOW I GOT THE ANSWER

It's one of the limits Chapter 23 puts on ESFR before you ever open a table. Screen first.

THE TRAP

If you said a four-foot curtain: you swapped the two numbers. The curtain is two feet deep. The aisle is four feet wide.

STEP B. CMSA.

5. FIND IT · BOOK OPEN **Our warehouse. Cartoned nonexpanded Group A plastic, double-row racks, thirty feet of storage, thirty-five-foot ceiling, CMSA. What head, how many design sprinklers, and what pressure?**

THE ANSWER

K-19.6 pendent, wet, fifteen design sprinklers, twenty-five psi.

WHERE I WENT

Chapter 22, Table 22.5, p. 13-222. Find the cartoned nonexpanded block, then the row with thirty feet of storage and thirty-five feet of ceiling. There's one option on that row.

HOW I GOT THE ANSWER

CMSA picked Chapter 22. Group A plastic in racks, up to thirty-five feet, picked Table 22.5. This table is keyed on storage height and ceiling height, so you need both numbers: thirty over thirty-five.

THE TRAP

If you said thirty psi: you read the thirty-five over forty row. That's thirty-five feet of storage under a forty-foot ceiling. Match both heights.

THE TRAP

If you said in-rack sprinklers required: you read the K-11.2 or K-16.8 upright rows at twenty-five feet of storage under a thirty-five-foot ceiling. Our storage is thirty feet.

STEP C. ESFR.

6. FIND IT · BOOK OPEN **Our warehouse. Thirty feet of storage under a thirty-five-foot ceiling, ESFR ceiling-only, K-25.2 pendent. What's the minimum pressure?**

THE ANSWER

Twenty psi.

WHERE I WENT

Chapter 23, Table 23.3.1, p. 13-226. Find the Class One through Four and cartoned nonexpanded Group A block. Go down to the thirty-five-foot ceiling row. Read across to the K-25.2 pendent column.

HOW I GOT THE ANSWER

ESFR picked Chapter 23, and ceiling-only picked this table. The table is keyed on ceiling height only. The thirty feet of storage never enters it.

THE TRAP

If you said fifteen psi: you read the thirty-foot row. Thirty is your storage height, and this table has no storage column. Use the thirty-five-foot ceiling row.

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7. FIND IT · BOOK OPEN **Same ceiling, thirty-five feet. Now the commodity is cartoned**

expanded plastic. What does ESFR ceiling-only give you?

THE ANSWER

Nothing. The thirty-five-foot row for cartoned expanded is all dashes. No ESFR ceiling-only.

WHERE I WENT

Chapter 23, Table 23.3.1, p. 13-226. Find the cartoned expanded block, then the thirty-five-foot ceiling row.

HOW I GOT THE ANSWER

The commodity picks the block, the ceiling picks the row. Cartoned expanded only has values at twenty-five and thirty feet of ceiling.

THE TRAP

If you said seventy-five or fifty-two psi: you read the Class One through Four and cartoned nonexpanded block. Expanded and nonexpanded are different blocks. Check the commodity before you read across.

STEP D. ESFR WITH ONE LEVEL OF IN-RACKS.

8. SAY IT · NO BOOK **Foam cups again, cartoned expanded plastic, thirty-five-foot ceiling. The ceiling-only table is all dashes. Can you add one level of in-racks and use the ESFR plus in-rack table?**

THE ANSWER

No. The ESFR plus in-rack table covers Class One through Four and nonexpanded plastics only, cartoned or exposed. Expanded plastic isn't in it.

WHERE I WENT

Table 25.5.4.1, p. 13-328.

HOW I GOT THE ANSWER

The in-rack option has its own commodity limits. A dash in the ceiling-only table doesn't open it for you. Separately, that table also caps the ceiling at forty-five feet.

THE TRAP

If you said yes: you heard "the ceiling-only table ran out" and stopped there. Check the commodity column of the ESFR plus in-rack table before you use it.

STEP E. HOSE AND DURATION.

9. SAY IT · NO BOOK **ESFR with twelve design heads. What's the hose allowance and duration? And standard-spacing CMSA with fifteen?**

THE ANSWER

ESFR, two hundred fifty gallons per minute, sixty minutes. CMSA, five hundred gallons per minute, ninety minutes.

WHERE I WENT

Table 20.15.2.6, p. 13-190.

HOW I GOT THE ANSWER

For CMSA and ESFR the hose table counts ceiling sprinklers in the design. Twelve falls in "up to twelve." Fifteen falls in "over twelve, up to fifteen."

THE TRAP

If you said five hundred and ninety minutes for ESFR: you put twelve heads in the over-twelve row. Twelve is "up to twelve."

Episode 6: In-Racks, Alternatives, Special Designs

9 answers. Cover everything under a question, say your answer and your path out loud, then uncover one line at a time.

WARM-UP

1. SAY IT · NO BOOK **Table 23.3.1, the ESFR ceiling-only table. What is it keyed on?**

THE ANSWER

The commodity picks the block, then the maximum ceiling or roof height picks the row. No storage height at all.

WHERE I WENT

Table 23.3.1, p. 13-226.

HOW I GOT THE ANSWER

The table has no storage-height column. You find the commodity, then the row for your ceiling height, then read across by K-factor.

THE TRAP

If you said storage height: that's the trap. Thirty feet of storage under a thirty-five-foot ceiling reads the thirty-five-foot row.

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2. SAY IT · NO BOOK **In an ESFR ceiling design, how many sprinklers are in the design area, and how are they laid out?**

THE ANSWER

The most demanding twelve sprinklers, four on each of three branch lines.

WHERE I WENT

§23.2.2.

HOW I GOT THE ANSWER

ESFR is designed by a head count, not a density over an area. And that count of twelve is what puts its hose at two fifty and sixty minutes.

THE TRAP

If you said fifteen: that's the CMSA head count for our warehouse, from Table 22.5.

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3. SAY IT · NO BOOK **CMSA with fifteen design heads. What hose stream and what duration?**

THE ANSWER

Five hundred gallons per minute, ninety minutes.

WHERE I WENT

Table 20.15.2.6.

HOW I GOT THE ANSWER

For CMSA and ESFR, that table is keyed to the number of ceiling sprinklers in the design. Fifteen sits in the over-twelve-up-to-fifteen row.

THE TRAP

If you said two fifty and sixty minutes: that's the up-to-twelve row. That's the ESFR answer, not CMSA.

STEP B. PICK THE ARRANGEMENT.

4. FIND IT · BOOK OPEN **Same cartoned Group A plastic over twenty-five feet, but you pick option two instead of option one. How far apart can the in-rack levels be, vertically?**

THE ANSWER

Fifteen feet.

WHERE I WENT

Chapter 25, option figure B, p. 13-314. Read the notes under the drawing. Note eight, the last one, gives the vertical spacing.

HOW I GOT THE ANSWER

"cartoned Group A over twenty-five feet" is the cartoned over-twenty-five section, which hands you to the option figures. "Option two" is part B, and each figure carries its own notes.

THE TRAP

If you said ten feet: you carried over option one. Ten feet is options one, four, five, six, and seven. And on that same page, note two puts face sprinklers in option two.

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5. SAY IT · NO BOOK **Our warehouse, option one, in-rack levels at ten and twenty feet only, thirty feet of storage. What's the ceiling density?**

THE ANSWER

Point four five over two thousand.

WHERE I WENT

The ceiling design box, printed the same on all seven option figures.

HOW I GOT THE ANSWER

Ten feet of storage sits above the top level. Point three zero is only for five feet or less. More than five, up to ten, is point four five.

THE TRAP

If you said point three zero: that's the answer only after you add the level at twenty-five feet.

STEP D. IN-RACKS UNDER ESFR.

6. FIND IT · BOOK OPEN **ESFR at the ceiling, open racks, one level of in-racks. The drawing shows K-5.6 standard-response in-racks. What has to change?**

THE ANSWER

Two things. The heads have to be quick-response, and at least K-8.0.

WHERE I WENT

Chapter 25, §25.5, p. 13-327. The ESFR in-rack head rule is on that page, and so is the five-foot horizontal spacing.

HOW I GOT THE ANSWER

The words "ESFR at the ceiling" pick §25.5, the in-racks-with-ESFR section. The CMDA option figures don't apply here.

THE TRAP

If you said the drawing is fine: you used the CMDA option figures. Their first note allows quick- or standard-response at K-5.6. Under ESFR, both of those are out.

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7. FIND IT · BOOK OPEN **Our warehouse: cartoned nonexpanded Group A, thirty feet of storage, thirty-five-foot ceiling, ESFR with one level of in-racks. The ceiling heads are K-16.8 pendent. How many ceiling heads, at what pressure?**

THE ANSWER

Twelve heads at sixty-three psi.

WHERE I WENT

Chapter 25, Table 25.5.4.1, p. 13-328. Find the K-16.8 pendent row and read across to the number of sprinklers and the pressure.

HOW I GOT THE ANSWER

"one level of in-racks" under ESFR means the one-level figure, and its first note sends the ceiling design to this table. Its rows are by K-factor.

THE TRAP

If you said fifty-two psi: you read Table 23.3.1, the ESFR ceiling-only table, at the thirty-five-foot ceiling row. That's the answer with no in-racks. If you said ninety: that's the K fourteen row.

STEP E. THE HYDRAULICS.

8. SAY IT · NO BOOK **The in-rack demand and the ceiling demand meet at the connection. What happens to the flows, and what happens to the pressures?**

THE ANSWER

The flows are added. The pressures are balanced to the higher of the two.

WHERE I WENT

§28.2.5.2, the balancing rule.

HOW I GOT THE ANSWER

Each side is calculated on its own first. Where they meet, the lower-pressure side gets raised to match.

THE TRAP

If you said the pressures add, or you split the difference: no. Flows add. Pressures go to the higher side, and the lower side doesn't get to pull it down.

---

9. FIND IT · BOOK OPEN **You're using §25.6, in-racks independent of the ceiling. Cartoned nonexpanded Group A plastic, option two B. What's the maximum vertical spacing**

between in-rack levels?

THE ANSWER

Forty feet.

WHERE I WENT

Chapter 25, Table 25.6.2.4.1, the independent spacing table, p. 13-336. Find the cartoned nonexpanded Group A row and read across to the option two B column.

HOW I GOT THE ANSWER

"independent of the ceiling" picks §25.6, not the CMDA option figures. This table is keyed on the commodity and the option.

THE TRAP

If you said thirty feet: you read the option two A column, or the cartoned expanded row, where option two B is thirty. If you said ten, fifteen, or twenty: those are the CMDA option figures, and this table doesn't use them.